

Practical DSP Tutorials & Implementation Guides

Step-by-step filter configurations, audio presets, stability auditing, and firmware code export.

Synthesis Methodology

Filter synthesis in Overtune 3 follows a 4-phase linear engineering workflow:

1. **Specification:** Define passband ripple δ_{pass} , stopband rejection δ_{stop} , sampling rate F_s , and corner frequencies (F_{c1}, F_{c2}) .
2. **Prototype Mapping:** Compute continuous-time analog poles and zeros, pre-warp frequencies via $\Omega = 2F_s \tan(\pi F_c / F_s)$, and execute bilinear substitution.
3. **Verification:** Audit pole-zero unit circle stability ($|p| < 1$), unwrapped phase, and group delay transit times.
4. **Deployment:** Export zero-allocation C99/C++20 source routines directly into production audio engines.

Standard Audio Presets & Implementation Recipes

Preset 1: Audio Sub-Bass Cut (30 Hz High-Pass)

- **Goal:** Eliminate turntable rumble, HVAC hum, and DC offset below 30 Hz without kick drum phase smearing.
- **Settings:** High-Pass, Butterworth, Order 4 (-24 dB/octave), $F_s = 48,000$ Hz, $F_c = 30$ Hz.
- **Audit:** Verify -3.0 dB at 30 Hz and monotonic roll-off exceeding -40 dB below 10 Hz.

Preset 2: AC Mains Hum Notch (50/60 Hz Band-Stop)

- **Goal:** Remove power supply ground-loop induction with > 50 dB rejection while preserving adjacent musical bass.
- **Settings:** Band-Stop, Elliptic or Chebyshev II, Order 4, $F_{c1} = 48$ Hz, $F_{c2} = 52$ Hz, $\delta_{\text{stop}} = 50$ dB.
- **Mechanism:** Places complex transmission zeros directly on the unit circle $z = e^{\pm j2\pi(50)/48000}$.

Preset 3: Voice Telephony Bandpass (300 Hz – 3.4 kHz)

- **Goal:** ITU-T G.712 speech channel isolation focusing human vocal formants and rejecting acoustic background rumble.
- **Settings:** Band-Pass, Chebyshev Type I, Order 4, 0.5 dB passband ripple, $F_{c1} = 300$ Hz, $F_{c2} = 3,400$ Hz.
- **Q-Factor:** $Q = f_0 / \text{BW} = \sqrt{300 \cdot 3400} / (3400 - 300) \approx 0.32$.

Preset 4: Anti-Aliasing Reconstruction Filter (20 kHz Lowpass)

- **Goal:** Brickwall cutoff for Hi-Res audio conversion, protecting Nyquist frequency at 48 kHz sampling rate.
- **Settings:** Low-Pass, Elliptic, Order 8, 0.1 dB passband ripple, 70 dB stopband rejection, $F_c = 20,000$ Hz.
- **Transition Band:** Drops from -0.1 dB at 20 kHz to -70 dB at 22 kHz.

Diagnostic Analysis & Stability Auditing

Before committing filter coefficients to firmware, review the diagnostic views in the **Analysis Suite** (Ctrl+2):

- **Pole-Zero Constellation:** Confirm all poles satisfy $\max_k |p_k| < 1.0$. Poles situated within $|p| > 0.985$ require 64-bit double precision state buffers to prevent coefficient quantization drift.
- **Group Delay τ_g :** High-order Chebyshev and Elliptic designs display localized group delay peaking at cutoff edges, producing transient dispersion. Use Bessel approximations where transient integrity is required.
- **Impulse & Step Responses:** Verify absence of sustained limit cycles, overflow saturation, or excessive overshoot ringing.

Embedded Firmware Export

Generate production deployment source in the **Production Code Exporter** (Ctrl+4):

- **Embedded C99:** Emits static coefficient tables `double b[N][3], a[N][3]` and a vectorized Direct Form II Transposed processing loop with zero heap allocation.
- **Modern C++20:** Generates a templated `BiquadCascade<N>` class with `constexpr` coefficient tables, SIMD auto-vectorization friendliness, and thread-safe sample execution.
- **Python SciPy:** Emits an executable verification script utilizing `scipy.signal.sosfilt` for automated testbench regression.

External References

- [Audio Filter Design Guide – Wikipedia](#)
- [Direct Form II Transposed Implementation – Wikipedia](#)
- [Low-Pass Filter Applications – Wikipedia](#)
- [High-Pass Filter Applications – Wikipedia](#)
- [Band-Pass Filter Applications – Wikipedia](#)
- [Overtune 3 Project Source Code Repository](#)